

LYON-BRON AP, France

WMO: 074800

Lat: 45.7261N

Lon: 4.9378E

Elev: 659

StdP: 14.35

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.5	25.6	14.6	11.8	28.1	19.0	14.6	28.9	29.9	46.2	26.7	46.7	5.4	0	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	92.9	68.4	89.3	67.7	86.1	66.9	70.8	86.7	69.4	84.4	68.2	82.5	11.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	97.6	75.6	64.4	92.9	74.8	63.1	88.7	74.0	34.9	86.5	33.8	84.8	32.8	82.8	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	24.7	21.2	18.5	DB	18.3	97.4	4.4	3.9	15.1	100.2	12.5	102.5	10.0	104.7	6.8	107.5
(5)				WB	17.4	73.3	4.4	1.7	14.2	74.6	11.6	75.6	9.2	76.6	5.9	77.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.5	39.0	41.1	48.0	54.4	61.2	68.9	72.7	71.3	64.6	56.8	46.6	40.6	(6)
(7)		DBStd	13.58	7.14	7.18	6.48	6.12	6.43	6.89	6.14	6.19	5.90	6.87	7.11	7.36	(7)
(8)		HDD50	1213	345	256	115	25	3	0	0	0	0	20	147	302	(8)
(9)		HDD65	4181	805	670	527	320	153	36	9	11	77	263	552	758	(9)
(10)		CDD50	3226	5	6	53	157	350	568	702	660	438	232	45	10	(10)
(11)		CDD65	720	0	0	0	3	35	155	246	207	65	9	0	0	(11)
(12)		CDH74	6866	0	0	0	48	333	1493	2435	1998	524	35	0	0	(12)
(13)		CDH80	2556	0	0	0	3	63	545	990	829	126	0	0	0	(13)
(14)	Wind	WSAvg	7.1	6.9	7.4	7.8	7.9	7.4	7.5	7.1	6.6	6.6	6.7	6.9	6.7	(14)
(15)	Precipitation	PrecAvg	33.5	2.1	1.7	1.9	2.9	3.1	2.7	2.7	2.6	3.6	4.1	4.0	2.1	(15)
(16)		PrecMax	44.3	5.3	4.3	5.7	7.6	7.0	6.4	6.0	6.1	10.0	10.4	11.1	6.2	(16)
(17)		PrecMin	25.8	0.6	0.2	0.2	0.4	0.7	1.0	0.8	1.0	0.9	0.7	1.3	0.4	(17)
(18)		PrecStd	5.0	1.1	1.0	1.2	1.7	1.5	1.3	1.3	1.2	2.3	2.0	2.3	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.4	62.3	70.5	79.4	85.6	95.2	97.2	99.2	87.9	77.8	67.1	61.1	(19)
(20)			MCWB	52.2	50.2	54.4	60.3	65.3	69.9	68.7	68.4	66.1	62.7	57.1	53.5	(20)
(21)		2%	DB	55.5	58.4	66.4	74.3	81.4	89.8	92.8	92.9	84.0	74.3	63.7	57.6	(21)
(22)			MCWB	49.4	49.2	52.4	57.7	62.8	68.2	68.6	68.2	65.7	61.9	55.9	51.2	(22)
(23)		5%	DB	52.8	55.0	62.9	70.6	77.7	86.1	89.1	88.6	80.2	71.1	60.8	55.0	(23)
(24)			MCWB	48.0	47.3	51.3	56.0	61.6	66.9	68.0	67.2	64.0	60.8	54.3	49.8	(24)
(25)		10%	DB	50.0	52.2	59.7	66.9	74.1	82.6	85.8	84.5	76.6	68.0	57.7	51.8	(25)
(26)			MCWB	46.2	46.0	50.1	54.0	60.0	65.6	67.0	66.6	63.0	59.5	52.4	47.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	51.6	56.3	61.1	67.5	73.1	72.5	71.8	69.7	65.1	58.4	53.9	(27)
(28)			MCD B	58.2	59.1	66.4	75.9	81.3	87.6	89.4	88.7	81.9	73.6	64.8	59.0	(28)
(29)		2%	WB	50.1	49.9	53.9	58.8	64.8	70.4	70.7	70.1	67.2	63.3	56.4	52.0	(29)
(30)			MCD B	54.5	56.7	62.8	72.2	78.0	85.2	87.4	85.7	78.8	71.4	62.2	57.1	(30)
(31)		5%	WB	48.4	48.2	52.5	56.8	62.9	68.4	69.4	68.9	65.6	61.7	54.8	50.1	(31)
(32)			MCD B	52.4	54.2	61.0	68.5	74.7	82.6	84.9	83.8	76.8	69.1	60.1	54.3	(32)
(33)		10%	WB	46.2	46.4	50.9	54.9	61.1	66.8	68.1	67.5	64.1	60.1	52.9	47.9	(33)
(34)			MCD B	49.8	51.7	58.7	65.0	71.1	80.0	82.7	81.4	74.4	66.7	57.4	51.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	13.5	17.0	18.1	18.3	19.6	19.7	20.3	18.8	15.5	12.0	10.5	(35)
(36)			MCD BR	14.5	19.5	23.0	24.9	24.5	24.8	25.1	27.3	25.0	19.5	16.6	14.5	(36)
(37)			MCWBR	10.0	12.0	12.0	11.7	10.2	9.4	8.5	9.2	9.9	9.6	9.9	9.7	(37)
(38)		5% WB	MCD BR	13.6	17.7	20.1	23.1	21.9	23.1	22.6	23.8	20.7	17.2	15.0	13.6	(38)
(39)			MCWBR	9.5	11.3	11.0	11.4	10.0	9.4	8.4	9.0	8.8	9.2	9.7	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.332	0.362	0.398	0.403	0.405	0.402	0.396	0.378	0.364	0.331	0.309	(40)	
(41)		taud	2.474	2.411	2.346	2.250	2.284	2.315	2.321	2.348	2.379	2.419	2.468	2.479	(41)	
(42)		Ebn at Noon	253	268	274	272	274	274	273	270	265	252	242	240	(42)	
(43)		Edh at Noon	22	28	34	41	41	40	39	36	33	27	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	396	665	1078	1434	1674	1994	1954	1688	1289	805	446	332	(44)	
(45)		RadStd	47	94	132	184	159	159	166	126	100	80	50	56	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page